PRAVAN - Chapter 1 Summary

Web foundations, illustrated

Programmation Avancée, Master 1 Informatique (Big Data), USTHB

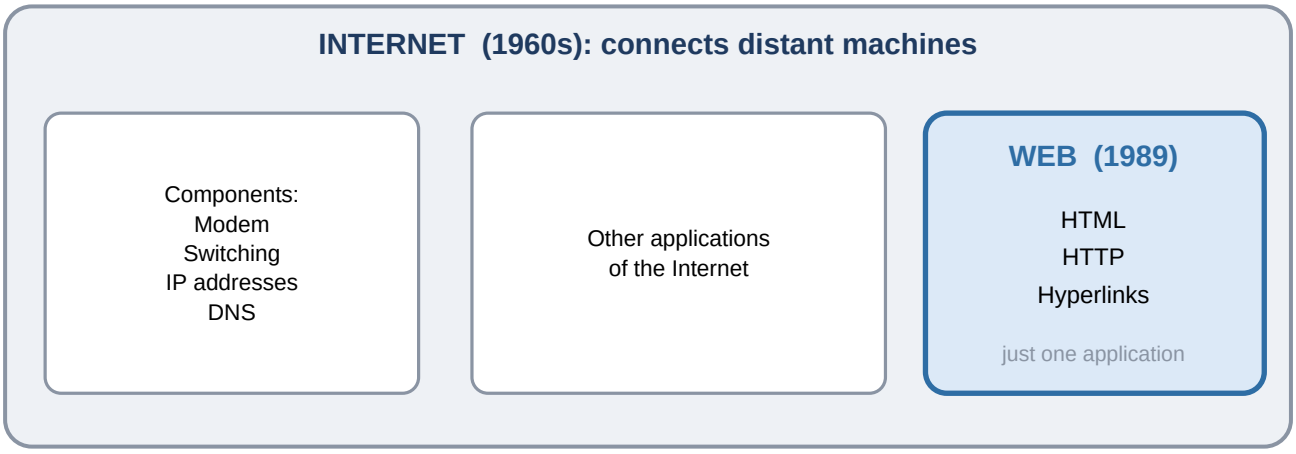
Why this course exists

The Big Data master trains people to support decision-makers. Data flows from sources, is extracted, and is shown in dynamic web pages (dashboards). PRAVAN teaches you to build those pages: a dynamic website with client-side and server-side features, using Python, JavaScript and web services.

PRAVAN covers the highlighted step: building the dynamic web pages

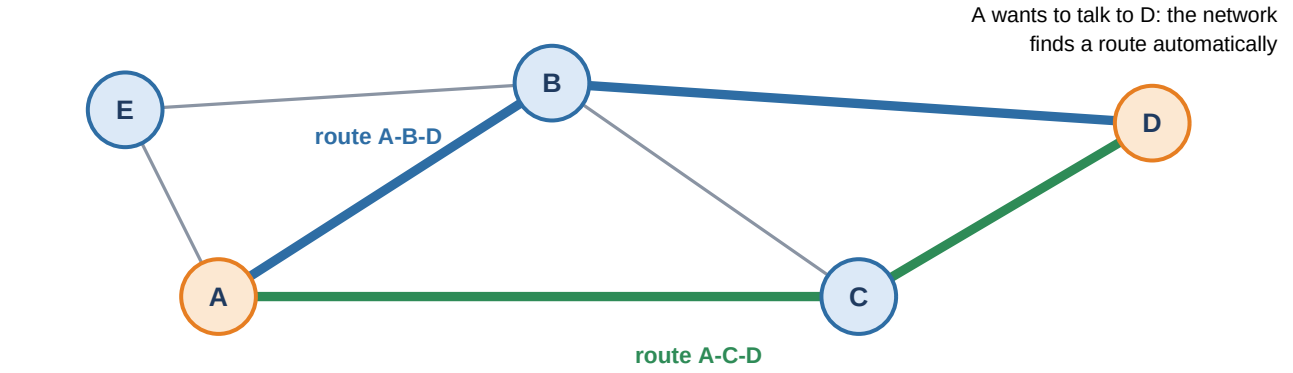
Internet ≠ Web

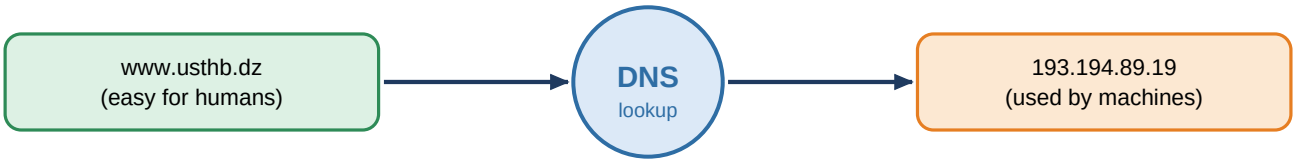
- **Internet (1960s):** the network that lets distant machines communicate.
- **Web (1989):** invented by Tim Berners-Lee at CERN to organize scattered information through hypertext (clickable links) and hypermedia (text, images, sound, video).
- The Web is **one application** of the Internet, not the Internet itself.



How the Internet works

- **Modem:** sends digital data over analog phone lines.
- **Automatic switching:** finds a route between machines, resiliently.
- **IP addresses:** unique ID per machine. **DNS:** maps human-friendly names to IPs.





IP format: XXX.XXX.XXX.XXX with each XXX from 0 to 255

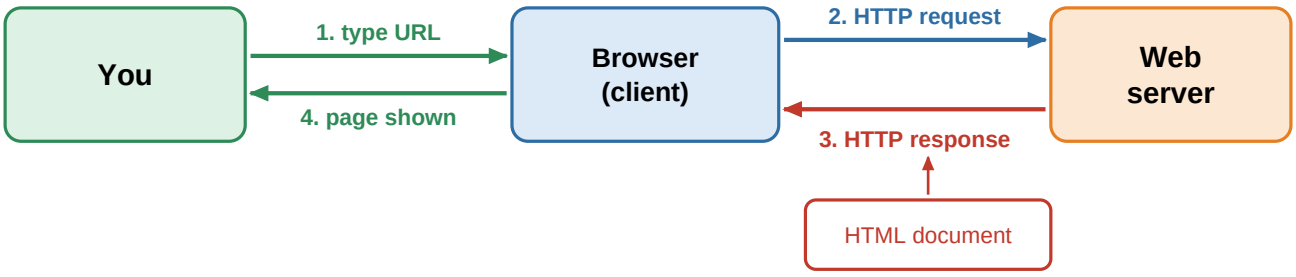
optional port after the address, e.g. 63.255.173.183:8080

How a page reaches you

Four steps, using the **HTTP** protocol: you type a **URL**, the browser sends a request, the **web server** replies with an **HTML document**, and the browser displays it.

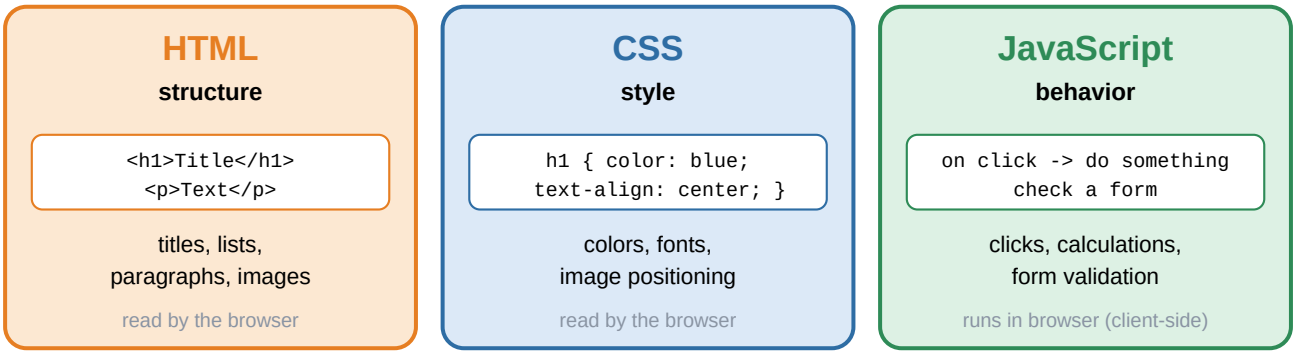
What happens when you type www.blender.org

The browser interprets the HTML it receives and draws the page (CTRL+U shows the HTML source)



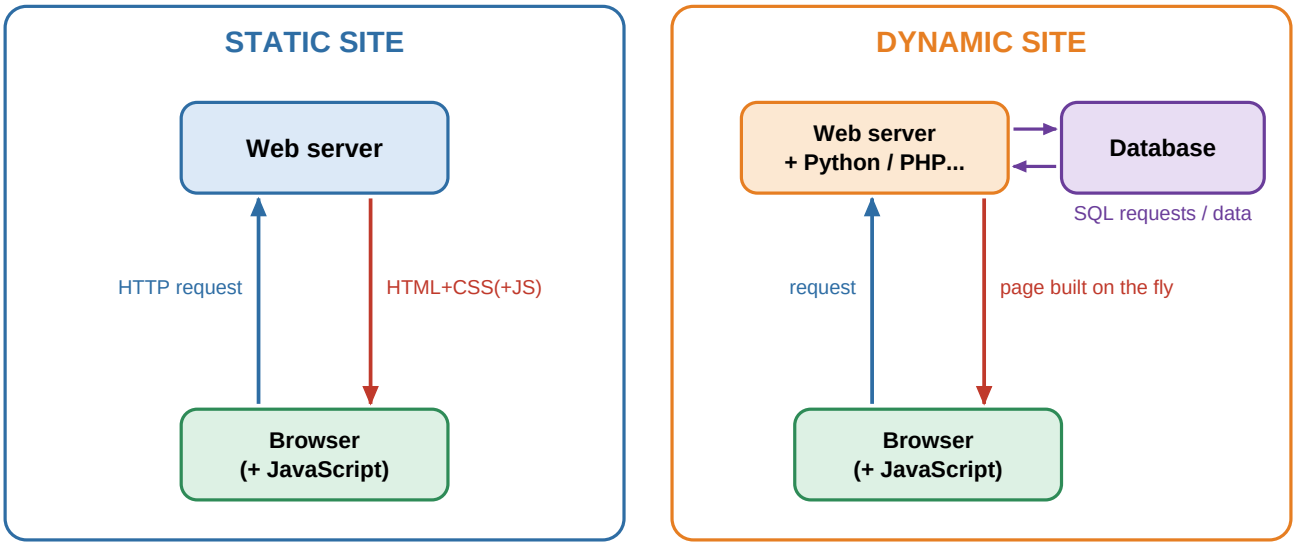
The building blocks of a web page

HTML gives the content its structure, CSS makes it look good, and JavaScript makes it interactive in the browser.



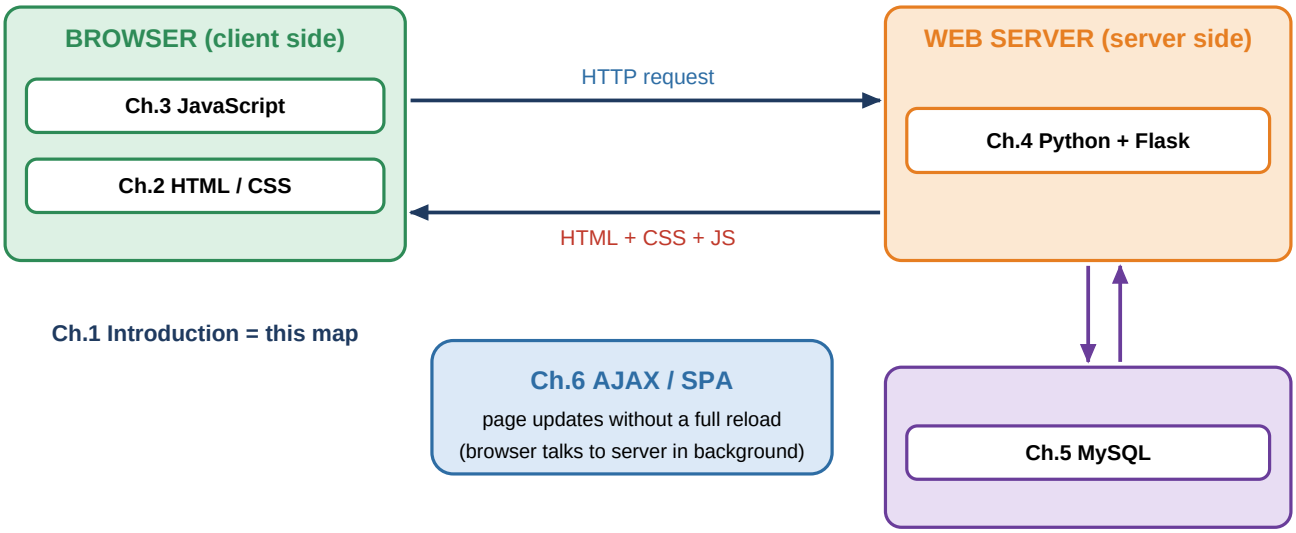
Static vs dynamic sites

- **Static:** the server always sends the same HTML+CSS (optionally with JavaScript for client-side interactivity).
- **Dynamic:** server-side code (Python, Java, PHP) builds each page on the fly based on the time, the user or a form, often using a **database** queried with SQL.



Course roadmap

Each chapter covers one part of the architecture.



Semester plan and grading

Semester: 11 weeks, then holidays before the evaluation (labs staggered between Info and Math groups)

Component	Mark
TP continuous assessment	/5
TP grade (project to submit, or test)	/15
TP total	/20
EMD (written exam)	/20
Credits / Coefficient	4 / 4